COMPUTER NETWORKS - UE24CS252B
Mininet Installation Manual on Existing Linux VM (Ubuntu)

Objective:

To install and configure Mininet on an already running Ubuntu-based virtual machine and verify its functionality for network simulation.

Introduction to Mininet:

Mininet is a virtual testbed or network emulator enabling the development and testing of network tools and protocols. It uses lightweight virtualization to make a single system look like a complete network, running the same kernel, system, and user code. With a single command, Mininet can create a realistic virtual network on any type of machine (Virtual Machine (VM), cloud-hosted, or native). Therefore, it provides an inexpensive solution and streamlined development running in line with production networks.

Mininet offers the following features:

- Fast prototyping for new networking protocols.
- Simplified testing for complex topologies without the need of buying expensive hardware.
- Realistic execution as it runs real code on the Unix and Linux kernels.
- Open source environment backed by a large community contributing extensive documentation.

Mininet's logical nodes can be connected into networks. These nodes are sometimes called containers, or more accurately, network namespaces. Containers consume sufficiently few resources that networks of over a thousand nodes have created, running on a single laptop. A Mininet container is a process (or group of processes) that no longer has access to all the host system's native network interfaces. Containers are then assigned virtual Ethernet interfaces, which are connected to other containers through a virtual switch. Mininet connects a host and a switch using a virtual Ethernet (veth) link.

Assumptions:

A virtual machine is already created using VMware / VirtualBox and is being used for all Mininet related experiments.

The VM is running a Linux OS (Ubuntu 20.04 / 22.04 preferred)

- User has:
  - Sudo privileges

- Internet connectivity inside the VM

System Preparation:

Before installing Mininet, ensure that the system is updated and ready. Run the following commands in a terminal:

```
sudo apt update
sudo apt upgrade -y
```

This ensures all packages are up to date and avoid any dependency issues.

Installing mininet

A. Method 1: Direct Installation (Simple and recommended)

Install Mininet using Ubuntu's package manager:

```
sudo apt install mininet -y
```

Note: In case **apt** install doesn't work, use **apt-get** instead.

This installs:

- Mininet
- Open vSwitch
- Required dependencies

B. Method 2: Installation from source (Optional and Advanced)

Step 1: Install dependencies

```
sudo apt install git build-essential python3-pip -y
```

Step 2: Clone repository

```
git clone https://github.com/mininet/mininet
cd mininet
```

Step 3: Run installation script

```
sudo ./util/install.sh -a
```

Verifying Installation

Starting Mininet:

```
sudo mn
```

You should see:

- A default topology created
- Mininet CLI prompt **(mininet>)**

Connectivity test:

Inside Mininet:

```
pingall
```

Expected result:

- All hosts successfully ping each other
- Output shows **0% packet loss**

Exit Mininet:

```
exit
```

```
zap@yabsolutely:~$ sudo mn --switch ovsbr --test pingall
[sudo] password for zap:
Sorry, try again.
[sudo] password for zap:
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 0.163 seconds
```

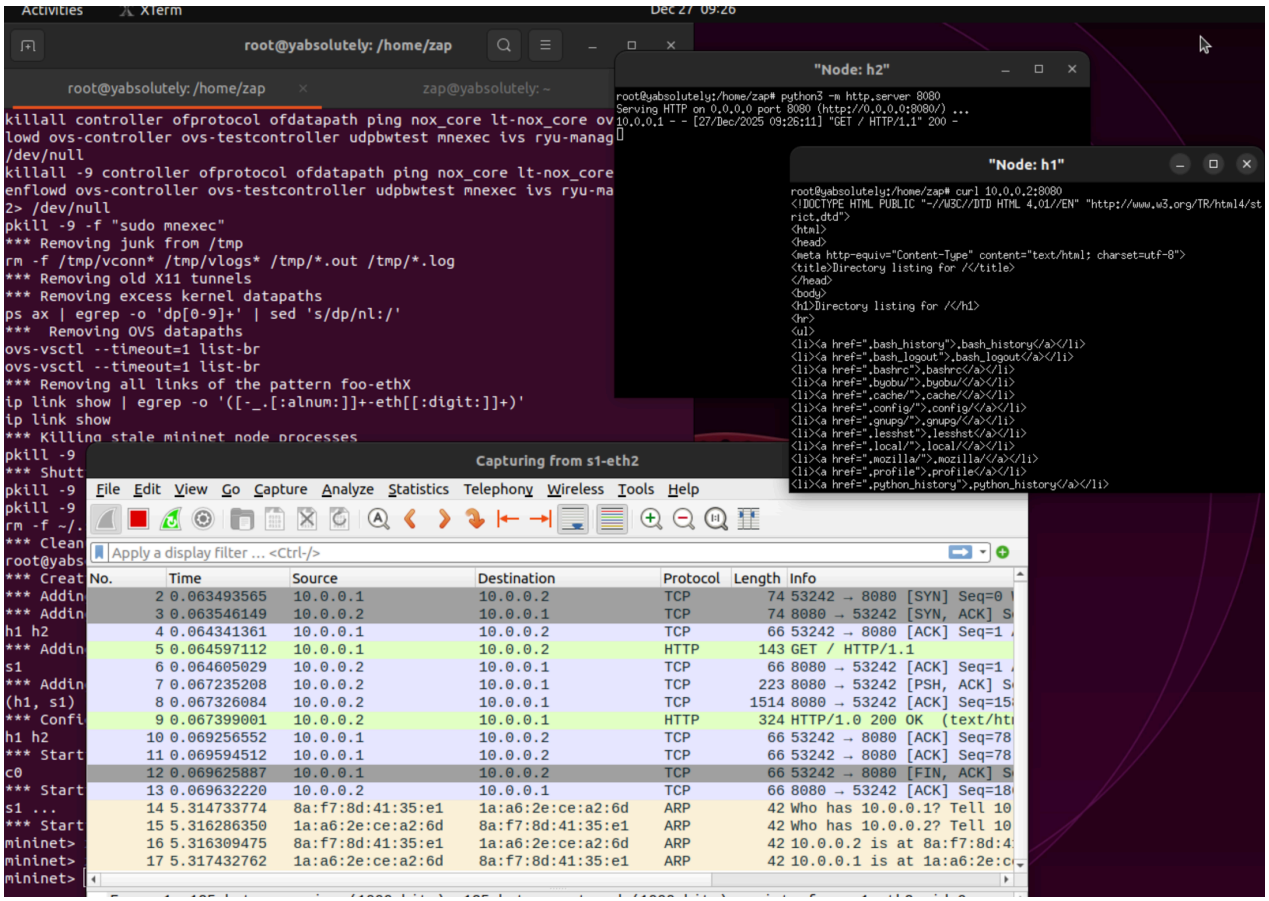
Running simple custom topology

Example: Single switch with 3 hosts

```
sudo mn --topo single,3
```

This creates:

- 1 switch
- 3 hosts



Important commands

Command	Purpose
`sudo mn`	Start default topology
`pingall`	Test connectivity
`nodes`	List nodes
`net`	View connections
`dump`	Node details
`exit`	Exit CLI

Cleanup and reset

To remove old configurations:

```
sudo mn -c
```

Trouble shooting

1. Command not found:

Reinstall Mininet:

```
sudo apt install mininet -y
```

2. Permission issues

Always run with sudo (**sudo mn**, then the rest as per manual).

3. Network issues within VM

Ensure:
VM network adapter is **NAT or Bridged**
Internet works (ping google.com)

4. Open vSwitch Errors

Install manually:

```
sudo apt install openvswitch-switch -y
```

References:

[1] Mininet overview - https://mininet.org/overview/
[2] https://mininet.org/walkthrough/
[3] https://github.com/mininet/mininet/blob/master/INSTALL
[4] https://github.com/mininet/mininet/wiki/Documentation